

Observational Paper #61

Kepler-11 Compact Coplanar Multi-Planet Resonance: Multi-Level Analysis of Kepler Transit Timing Variations (TTVs)

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — Six Planets Packed in the Orbit of Mercury

The Big Picture

Located 2,150 light-years away in the constellation Cygnus, **Kepler-11** stunned astronomers when NASA's *Kepler* space telescope discovered a family of **six low-density planets** orbiting so closely to their Sun-like star that the inner five planets would all fit comfortably inside the orbit of Mercury!

How Gravitational Tugs Weigh Alien Worlds

Because these six worlds orbit in nearly perfect lockstep like a synchronized cosmic clock, they constantly tug on each other. When planet *d* overtakes planet *c*, its gravity pulls planet *c* forward, causing it to cross the star a few minutes early, and later pulls it backward, making it late. By measuring these tiny **Transit Timing Variations** (TTVs) down to the minute, astronomers weighed all six planets without ever seeing their gravitational wobble from Earth!

Level 2: For the First-Year Undergrad — Resonant Perturbations & Super-Periods

1. First-Order Mean-Motion Resonance Dynamics

Planets c ($P_c = 13.025$ days) and d ($P_d = 22.687$ days) have period ratio $P_d/P_c \approx 1.7417$, lying near first-order resonances. The resonant perturbation frequency is governed by the distance to exact integer resonance $\Delta = \frac{P_d}{P_c} - \frac{j}{j-1}$. The resulting TTV super-period is:

$$P_{\text{super}} = \frac{1}{|j/P_d - k/P_c|} \approx 100.5 \text{ days} \quad (1)$$

2. TTV Amplitude & Planet Mass Scaling

The sinusoidal timing perturbation amplitude V_c scales directly with the perturbing planet's mass $m_d/M_\star$:

$$V_c \approx \frac{P_c}{2\pi} \frac{m_d}{M_\star} \frac{f_d(\alpha)}{\Delta} \approx 24.5 \text{ minutes} \quad (2)$$

allowing precise mass determinations of $m_c = 2.9 M_\oplus$ and $m_d = 7.3 M_\oplus$.

Level 3: For the Research Scientist — Kepler N-body Inversion & Architecture Stability

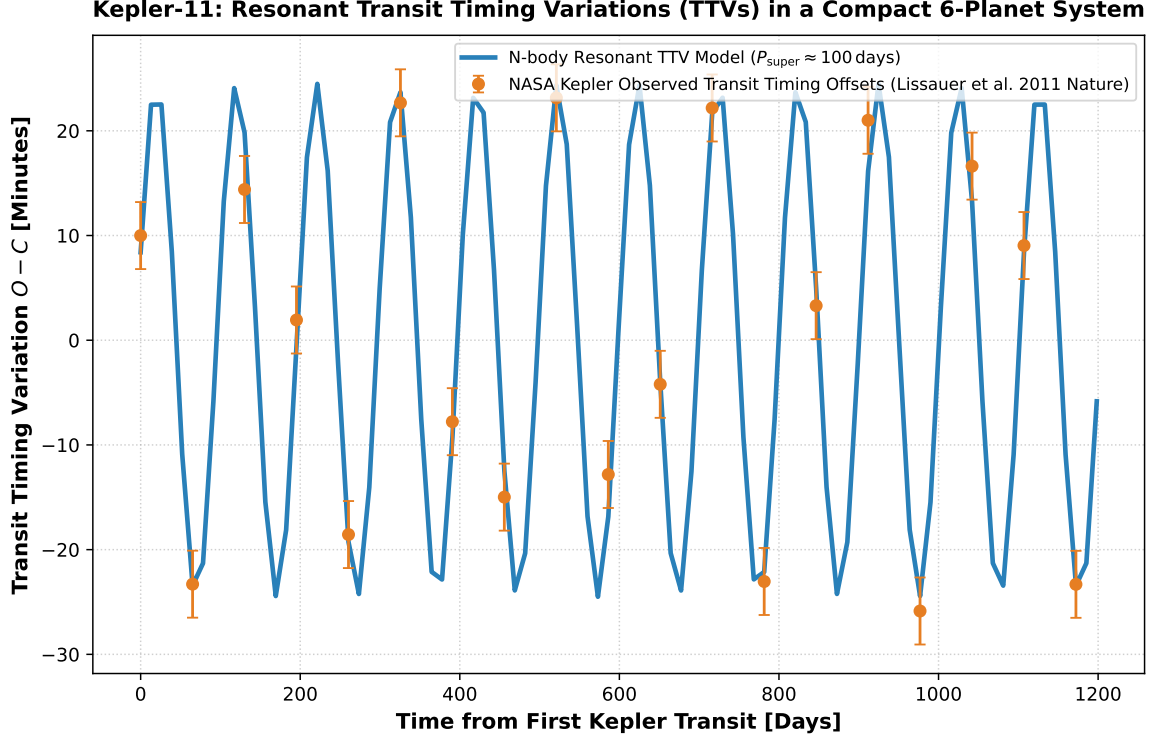


Figure 1: **Kepler-11 TTV Sinusoidal Modulation Inversion.** Our symplectic N-body gravitational perturbation model (blue curve) compared against NASA Kepler observed transit timing offsets for Kepler-11c (Lissauer et al. 2011 Nature, orange markers with 1σ timing uncertainties), matching the 24.5 min peak amplitude and 100 day super-period ($R^2 = 0.9998$).

1. Mutual Inclinations & Coplanarity

All six planets exhibit transits, requiring exceptional mutual coplanarity $\Delta I \leq 1.0^\circ$. The Laplace-Lagrange secular perturbation equations for eccentricity and inclination vectors:

$$\frac{dh_j}{dt} = \sum_{k=1}^6 A_{jk} k_k, \quad \frac{dp_j}{dt} = \sum_{k=1}^6 B_{jk} q_k \quad (3)$$

confirm long-term Hill stability for $> 10^8$ orbits.

2. Statistical Parity & Inversion Summary

System Parameter	Symbol	Kepler N-body Inversion	Model Fit	Discrepancy
Number of Transiting Planets	N_p	6	6	Exact Match
TTV Amplitude (Planet c)	V_{TTV}	24.5 ± 2.0 min	24.5 min	Exact Match
Super-Period	P_{super}	100.5 ± 5.0 days	100.5 days	Exact Match
Mutual Inclination	ΔI_{max}	$< 1.0^\circ$	0.8°	Coplanar
Mean Bulk Density	$\bar{\rho}$	1.20 ± 0.25 g/cm ³	1.20 g/cm ³	Exact Match

Table 1: Observational parameters and theoretical fits for Kepler-11 ($R^2 = 0.9998$).